

Pranav Reddy

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PROJECTS & EXPERIENCE

Quantum Codesign Lab

June 2025 — present

Niu Group, UCSB Undergraduate Researcher

- Designed time-dependent Lyapunov control schedules with learnable parameters to accelerate ground state convergence in the Transverse Field Ising Model (TFIM) via feedback-based quantum algorithms (FQAs)
- Benchmarked GRU, Bayesian, and threshold decoders on continuous stabilizer readout across four hardware-realistic simulation phases; GRU achieved 96% accuracy under colored noise, measurement backaction, and drift

Pando.AI

July 2024–January 2025

Engineer Intern

<https://pando.ai>

- Built a real time retrieval-augmented automation pipeline with LLM-based inference to supply chain workflows
- Developed scalable alerting infrastructure and orchestration logic to improve response reliability and operational throughput in production enterprise environments

TensorDescent

January 2026 - April 2026

Language: Python

- Built multi-stage ML compiler pipeline using PyTorch FX by implementing ShapeProp, DCE, and operator fusion passes, reducing op count and eliminating intermediate memory roundtrips.
- Designed a lifetime-based memory planner for on-chip buffer reuse, mirroring memory tradeoff constraints

KernelForge

February 2026 - April 2026

Language: Python

- Built a Triton-inspired kernel compiler that lowers tensor ops through loop tiling (32x32) and CUDA-style codegen, mapping work across GPU thread hierarchies via block/threading indexing
- Implemented fusion and tiling optimization passes that eliminate kernel launch overhead and keep intermediate tensors in shared memory rather than global DRAM

AccelSim

February 2026 - April 2026

Language: Python

- Built a cycle accurate neural accelerator simulator modeling 5-instruction ISA with configurable latency, on-chip buffer management, and memory traffic tracking
- Implemented bottleneck detection classifying workloads as compute-bound, memory-bound, or buffer limited

Real-Time Quantum Error Decoding (3-Qubit Bit-Flip Code)

January 2026–Current

Language: Python

- Simulated continuous noisy measurement signals from a 3-qubit error-detecting code, modeling hardware realistic error processes for fault-tolerant quantum systems
- Developed decoding pipelines to identify and correct bit-flip events in real time, benchmarking learned models against optimal Bayesian inference baselines

- Engineered scalable experimental infrastructure to evaluate performance across stochastic noise regimes, emphasizing latency, reliability, and hardware-constrained signal decoding
- Explored sequence based and graph based ML approaches (RNNs, GNNs) to improve error classification accuracy under compute constraints

Application of Graphs to Machine Learning

January 2024–March 2024

Language: C++

- Implemented graph algorithms (BFS, DFS) to traverse and manage neural network structures, developed methods for node and connection management and applied graph traversal algorithms for prediction and back-propagation

EDUCATION

BS Electrical and Computer Engineering | University of California, Santa Barbara

September 2023 — Present

Pursuing electrical engineering major with an emphasis on Photonics, ML System Accelerators, High Performance Computing, Computer Architecture, Quantum Error Correction, ML/AI optimizations.

COURSE WORK

Intro to Photonics (ECE136A)	Probability and Stats (ECE139)	Computer Architecture (ECE152)
Quantum Photonics (ECE 136C)	Intro to Fields & Waves (ECE134)	Fiber Optic Systems (ECE 135)
Intro Solid State Devices (ECE132)	Intro to Optimizations (ECE 133)	Digital Circuit Design (ECE 10)
Problem Solving in C++ (CS 24)	Signal Processing (ECE 130A)	Discrete Processing (ECE 130B)
Interface Design (ECE153B)	Digital Design (ECE152)	Waveguide Physics (ECE 144)

SKILLS & LEADERSHIP

Languages: C++, C, Python, Java, Arduino, LATEX, SystemVerilog, low-level Linux
Tools: Unix, Git/Github, LangChain, Tavily, OpenAI, Claude, Gemini, Quartus, Logisim, FPGA
Sigma Phi Epsilon - Vice President / Housing Manager / Siblinghood Chair September 2023 - present